Reachability Analysis Based Localization of Limit Cycle in Benchmark Planar Oscillators

Sidhanta Mohanty[1] *, **Shaunak Sen**[2] *

* *Department of Electrical Engineering, Indian Institute of Technology Delhi, New Delhi, India (e-mail:* [1] *sidhanta.mohanty@ee.iitd.ac.in,* [2] *shaunak.sen@ee.iitd.ac.in).*

Abstract: Oscillatory dynamics are fundamental to systems in engineering, physics, and biology. In planar systems, the Poincaré-Bendixson theorem provides a geometric criterion for the existence of periodic orbits. Various analytical and geometrical approaches have been used for higher-order systems to obtain conservative and large region of existence of oscillatory behaviour. However localization of the limit cycle trajectory in these regions is desirable. In this work, we propose a interval-based reachability analysis to localize the limit cycle trajectory in the region of existence. Instead of propagating individual trajectories, we evolve hyperrectangular sets of initial conditions forward in time and examine the geometric relationship between initial and reachable sets to characterize oscillatory behavior. As a proof of concept, the framework is applied to benchmark planar oscillators like the Van der Pol oscillator and the Glycolysis model, demonstrating feasibility and establishing a computational foundation for future extensions to higher-dimensional nonlinear oscillators.

Keywords: Limit Cycle, Reachability Analysis, Van der Pol Oscillator, Glycolysis.

1. INTRODUCTION

Oscillatory dynamics are essential to the functioning of complex systems in engineering, physics, and biology. Critical biological processes like the cell cycle and circadian rhythm depend on periodic behavior to maintain optimal performance (Alon, 2019). Periodic behaviour are also crucial for various technical applications, such as quartz crystal oscillators utilized for precise timing in communication devices.

Given their significance, determining the existence of limit cycles in various systems has garnered interest of researchers. In planar systems, the Poincaré-Bendixson theorem provides a geometric criterion for the existence of periodic orbits. (Khalil, 2002). The fundamental concept of this theorem is to construct a positively invariant set containing no steady state points so that the trajectories eventually converge to a periodic orbit. However, its application is restricted to planar autonomous systems, and extensions to higher dimensions has garnered interest of researchers. Subsequently, several works have now provided various analytical and geometrical approaches to prove the existence of limit cycles in these higher-order systems (Tyson, 1975; Hastings et al., 1977; Hori et al., 2011; Mohanty et al., 2023; Katz et al., 2025. Similar to the Poincaré-Bendixson theorem, these approaches also establish the existence of a limit cycle without an explicit analytical solution. The region of existence of limit cycle obtained from these methods are large in size. This leads to the need of localizing the region of existence where the limit cycle trajectory lies.

Recent works in Interval Analysis have focused on the tighter enclosures of solution sets. An Interval Analysis based strategy has been developed to compute outer approximations of all invariant sets of a continuous-time non-linear dynamic system which are included inside a prior set of the state space via the use of constraint propagation (Le Mézo et al., 2017). While a limit cycle is technically an invariant set, the strategy is not designed to localize limit cycle trajectory within the phase space. The Set Inversion Via Interval Analysis (SIVIA) framework combined with Monte Carlo method has also been used to obtain validated enclosures in case of Van der Pol oscillator. (Mahato et al., 2019). However, the above methods face problems of scalability and computational complexity when applied to systems with higher dimension ($n \geq 3$).

To address this gap, we present a preliminary interval-based reachability analysis framework for localizing the region of the phase space where the limit cycle trajectory lies. Instead of tracking single trajectories, we propagate hyperrectangular sets of initial conditions forward in time and analyze whether their reachable sets return to, overlap with, or does not intersect with the initial set. This enables a geometric characterization of the oscillatory behavior. While the long-term objective is to extend this methodology to higher-dimensional nonlinear oscillators including biomolecular oscillators like the repressilator, this paper focuses on demonstrating the feasibility in a planar setting. As a proof of concept, we apply the proposed approach to benchmark planar nonlinear systems exhibiting stable limit cycles, thereby establishing the conceptual and

computational foundation for future extensions to higher dimensional systems.

2. PROBLEM STATEMENT

Consider a continuous-time nonlinear dynamical system

$$\dot{x}(t) = f(x(t)), \quad x(t) \in \mathbb{R}^2. \tag{1}$$

Let $\mathcal{X}_0 \subset \mathbb{R}^2$ denote a set of initial conditions. The objective is to determine whether the forward-reachable set

$$\mathcal{X}(t) = \{x(t) \mid x(0) \in \mathcal{X}_0\}, \tag{2}$$

exhibits traversal behavior consistent with the presence of a limit cycle. Specifically, we seek to classify the initial region $\mathcal{X}_0$ into one of the following categories:

- **Full containment:** The reachable set returns to and is a subset of $\mathcal{X}_0$, indicating that a limit cycle is fully enclosed.
- **Partial intersection:** The reachable set intersects $\mathcal{X}_0$ without complete enclosure, indicating partial overlap with a limit cycle.
- **Exclusion:** The reachable set does not intersect $\mathcal{X}_0$, indicating the absence of oscillatory behavior in that region.

3. METHODOLOGY

The methodology involves discretizing an initial phase space X_{search} into a collection of compact sets of initial condition. For each set of initial condition X_0, system dynamics are evolved forward in time using interval-based reachability analysis (Forets et al., 2019), producing an over-approximation of the flowpipe that encloses all trajectories originating within X_0. From this flowpipe, a return set X_t is extracted over a prescribed time horizon. The classification of X_0 is then determined by analyzing the geometric relationship between the return set X_t and the original region X_0. Each region is assigned to one of the following categories: (i) Completely contains the limit cycle, (ii) Partially contains the limit cycle, or (iii) Does not contain the limit cycle.

The containment is evaluated using a directional slicing test, in which one-dimensional cross-sections are swept across the plane to determine whether the projected return set is geometrically enclosed within the projected initial set. The two-dimensional setting enables direct visualization of reachable sets and classification outcomes, allowing validation of the framework before extending it to higher-dimensional oscillatory systems.

4. ALGORITHM

Algorithm 1 presents an interval-based reachability analysis framework for identifying and localizing the region of the phase space that contains the limit cycle trajectory of a planar oscillators.

The search space is partitioned into smaller rectangular interval subsets $\mathcal{S}$ using `divide`. This returns a collection of rectangular boxes X_0. For each subset $X_0 \in S$ as an initial condition, the system dynamics are evolved using interval-based reachability analysis. Here, the initial condition is treated as an interval set, and the solution is computed as a flowpipe over time. This produces a sequence of reachable sets that overapproximate all trajectories originating from X_0. To determine whether trajectories exhibit oscillatory behavior, a return set X_t within the computed flowpipe

Algorithm 1 Reachability Analysis for Limit Cycle Localization with various functions in an hierarchical order.

Require: Search space X_{search}, Number of subdivisions n_{div}, skip parameter s, number of directions N
Ensure: Partition of X_{search} into classified subsets

▷ **Initialize output lists:**
$Limit_Cycle \leftarrow [\,]$, $idx_limit \leftarrow [\,]$,
$Limit_Cycle_partial \leftarrow [\,]$, $idx_partial \leftarrow [\,]$,
$Further_check \leftarrow [\,]$, $idx_further_check \leftarrow [\,]$,
$Failed_Boxes \leftarrow [\,]$

▷ **Divide search space into smaller intervals**
$S \leftarrow$ `divide`(X_{search}, n_{div})

▷ **Iteratively solve each initial condition set**
for all $(X_0, idx) \in S$ **do**

  ▷ **Evolve system dynamics**
  $\mathcal{S} \leftarrow$ `solve_ivp`(X_0)

  ▷ **Determine closest return set**
  $X_t \leftarrow$ `nearest`$(\mathcal{S}, X_0, s)$
  if $X_t =$ failure **then**
    Append X_0 to $Failed_Boxes$
  end if

  ▷ **Perform containment test**
  $status \leftarrow$ `classify_two_planes`(X_0, X_t, N)
  if $status =$ "Contains limit cycle completely" **then**
    Append X_0 to $Limit_Cycle$
    Append idx to idx_{limit}
  else if $status =$ "Contains limit cycle partially" **then**
    Append X_0 to $Limit_Cycle_partial$
    Append idx to $idx_{partial}$
  else if $status =$ "Needs further check with stricter settings" **then**
    Append X_0 to $Further_Check$
    Append idx to $idx_{further}$
  end if
end for
return output lists

Subroutine: classify_two_planes(X_0, X_t, N)
  calls `multi_line_check`(X_0, X_t, N)
  return containment status

Subroutine: multi_line_check(X_0, X_t, N)
  Generate N directions $\{u_k\}$
  for each u_k:
    calls `hull_line`(X_0, u_k)
    perform geometric containment check
  return directional verification result

Subroutine: hull_line(Z, u)
  compute extreme points via support function σ
  return boundary line segment

is identified using `nearest`, such that it is the reachable set nearest to the initial set X_0 after a prescribed time horizon.

After obtaining X_t, the containment is verified in $\mathbb{R}^2$ using `classify_two_planes`. This subroutine determines whether the return set X_t is geometrically enclosed within the initial set X_0. The function `multi_line_check` generates N uniformly distributed directions. For each direction u_k, the subroutine `hull_line` computes the extreme points of a zonotopic set Z using the support function σ. The support function determines the farthest point of the set in direction u_k and in the opposite direction $-u_k$, defining the corresponding boundary line segment. In each direction, containment is verified by checking

$$\sigma_{X_t}(u_k) \leq \sigma_{X_0}(u_k) \quad \text{and} \quad \sigma_{X_t}(-u_k) \leq \sigma_{X_0}(-u_k).$$

If these inequalities hold for all N directions, the return set is considered geometrically enclosed within the initial set.

Based on containment results, each subset X_0 is categorized as:

- **Contains limit cycle completely:** Containment check hold in all direction signifying complete enclosure of return set X_t within initial set X_0.
- **Contains limit cycle partially:** Partial containment is observed.
- **Needs further check:** Cases requiring stricter numerical settings.

The index of each subset is stored alongside its classification. After iterating over all subsets, the algorithm returns the classified partition of the original search space. The subsets marked as completely or partially containing the limit cycle together gives a rigorous localization of the region where the periodic orbit lies in the phase space.

5. CASE STUDIES

We use the following benchmark planar oscillators to establish the efficacy of the interval-based reachability analysis discussed in the previous sections.

5.1 Van der Pol Oscillator:

The differential equations describing the dynamics of the Van der Pol oscillator are given below (Khalil, 2002).

$$\begin{aligned} \dot{x_1} &= x_2, \\ \dot{x_2} &= -x_1 + \mu(1 - x_1^2)x_2, \end{aligned} \tag{3}$$

where $\mu > 0$ is the parameter that determines the nonlinearity and the damping of the system. For the simulations in the later sections, the value of μ is taken as 1.5.

5.2 Glycolysis Model:

The differential equations describing the dynamics of the Glycolysis model are given below (Strogatz, 2024).

$$\begin{aligned} \dot{x_1} &= -x_1 + ax_2 + x_1^2 x_2, \\ \dot{x_2} &= b - ax_2 - x_1^2 x_2, \end{aligned} \tag{4}$$

where $a, b > 0$ are kinetic parameters. For the simulations in the later sections, the value of a and b are taken as 0.02 and 0.5 respectively.

6. RESULTS

The interval-based reachability-analysis successfully classifies regions of the Van der Pol oscillator's phase space according to its dynamical behavior. In Figure 1, the partitioned region containing the system's steady state $(0, 0)$ is excluded from the reachability analysis. The partitioned region near the excluded steady-state point is classified as "Completely contains limit cycle (LC)". For trajectories initialized in the region near the steady state, the reachable sets contract monotonically as they approach equilibrium. Consequently, the final contracted overapproximation is geometrically contained within the initial, larger zonotope, satisfying the containment criterion despite the absence of periodic behaviour. A distinct cluster of initial sets, in Fig. 1, was classified as "Completely contains LC" or "Partially contains LC". This region is located far from the system's unstable steady state. Further the numerical solution of the Van der Pol oscillator with initial condition $(2, -1)$ is plotted. The trajectory passes through the region classified by the method as containing the limit cycle completely or partially.

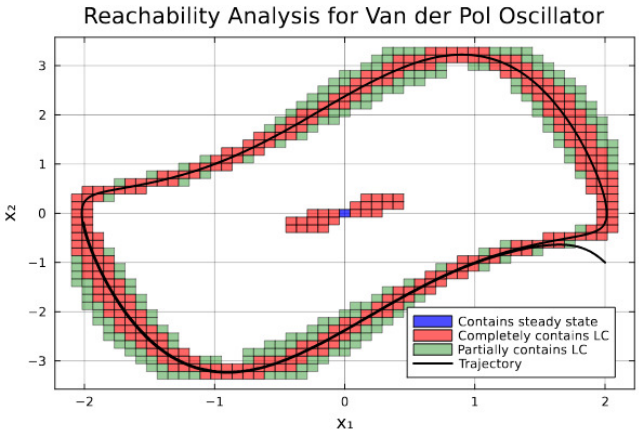


Fig. 1. Utilizing the reachability analysis on the Van der Pol oscillator, the search space $[-2.1, 2.1] \times [-4, 4]$ is classified into regions (a) Completely containing LC and (b) Partially containing LC. The numerical solution with initial condition $(2, -1)$ passes through the region marked as Completely containing LC. The system parameter used in the simulation is: $\mu = 1.5$.

We also use the interval-based reachability analysis to classify the phase space of the Glycolysis model according to its dynamical behavior. This is visually summarized in Fig. 2. A cluster of initial sets was classified as "Completely contains LC" or "Partially contains LC" through which the numerical solution with initial condition $(0.27, 4.9)$ passes through. A cluster of initial sets near the steady set is falsely marked 'Completely contains LC" due to the the reachable sets contracting monotonically as they approach equilibrium.

The classification of the phase space for the benchmark planar oscillators using the interval-based reachability analysis validates the efficacy of the framework in localizing the region in which the limit cycle trajectory lies.

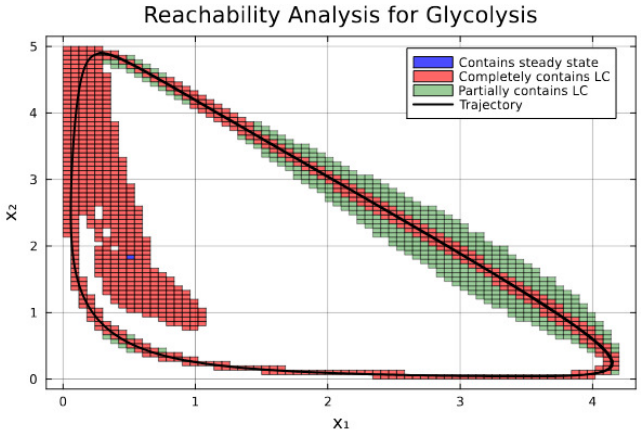


Fig. 2. Utilizing the reachability analysis on the Glycolysis model, the search space $[0, 4.5] \times [0, 5]$ is classified into regions (a) Completely containing LC and (b) Partially containing LC. The numerical solution with initial condition $(0.27, 4.9)$ passes through the region marked as Completely containing LC. The system parameters used in the simulation are: $a = 0.02$ and $b = 0.5$.

7. CONCLUSION AND FUTURE WORK

We present an interval-based reachability analysis to identify and localize the regions of the phase space where the limit cycle trajectory of the nonlinear oscillator lies. As a proof of concept, we applied the reachability analysis framework to benchmark planar oscillators - the Van der Pol oscillator and the Glycolysis model. We used the framework described in Algorithm 1 to classify the search space into regions marked as "Completely contains LC" and "Partially contained LC". For initial conditions close to the steady state, the reachable sets contract monotonically toward equilibrium. The final overapproximation may lie inside the first zonotope, which means it passes the inclusion test even if it does not exhibit periodic behavior. Subsequently, interpreting the false complete classifications near the steady state due to contraction is also essential for validating the overall analysis. So, differentiating the limit cycle containment from contraction-induced inclusion is therefore an important step in the reachability analysis framework.

Although the present implementation focuses on planar dynamical systems, the framework is designed to generalize to higher-dimensional oscillators. In future extensions, the primary modification will be concentrated in the function `classify` and `classify_two_planes`, which will be expanded to accommodate multi-dimensional zonotopic reachable sets through multi-plane projection and higher-dimensional geometric subset testing. The remaining components of the algorithm remain structurally unchanged, thereby supporting extension to complex oscillatory systems, such as cyclic gene regulatory networks and other high-dimensional nonlinear biological oscillators.

REFERENCES

Alon, U. (2019). *An introduction to systems biology: design principles of biological circuits*. Chapman and Hall/CRC.

Forets, M., Frehse, G., Potomkin, K., and Schilling, C. (2019). Juliareach: A toolbox for set-based reachability. *Proceedings of the 22nd ACM International*.

Hastings, S., Tyson, J., and Webster, D. (1977). Existence of periodic solutions for negative feedback cellular control systems. *Journal of Differential Equations*, 25(1), 39–64.

Hori, Y., Kim, T.H., and Hara, S. (2011). Existence criteria of periodic oscillations in cyclic gene regulatory networks. *Automatica*, 47(6), 1203–1209.

Katz, R., Giordano, G., and Margaliot, M. (2025). Instability of equilibrium and convergence to periodic orbits in strongly 2-cooperative systems. *Journal of Differential Equations*, 444, 113651.

Khalil, H. (2002). *Nonlinear Systems*. Pearson Education. Prentice Hall.

Le Mézo, T., Jaulin, L., and Zerr, B. (2017). An interval approach to compute invariant sets. *IEEE Transactions on Automatic Control*, 62(8), 4236–4242.

Mahato, N.R., Jaulin, L., Chakraverty, S., and Dezert, J. (2019). Validated enclosure of uncertain nonlinear equations using sivia monte carlo. In *Recent Trends in Wave Mechanics and Vibrations: Select Proceedings of WMVC 2018*, 455–468. Springer.

Mohanty, S., Mohit, A., Aggarwal, M., and Sen, S. (2023). An elementary proof for the existence of limit cycle in three gene cyclic regulatory network. In *2023 Ninth Indian Control Conference (ICC)*, 401–406. IEEE.

Strogatz, S.H. (2024). *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Chapman and Hall/CRC, 3rd edition.

Tyson, J.J. (1975). On the existence of oscillatory solutions in negative feedback cellular control processes. *Journal of Mathematical Biology*, 1(4), 311–315.